



Convection heat transfer from discrete heat sources in a liquid cooled rectangular channel

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Abstract

Steady-state experiments are performed to study general convective heat transfer from an in-line four simulated electronic chips in a vertical rectangular channel using water as the working fluid. The experimental data covers a wide range for laminar flow under natural, mixed and forced convection conditions with Reynolds number based on channel hydraulic diameter ranging from 40 to 2220 and Reynolds number based on heat source length ranging from 50 to 2775, respectively. The heat flux ranges from 0.1 W/cm² to 0.6 W/cm². The inlet water temperature is at 24 °C. The effect of heat fluxes, flow rates and geometrical parameters such as chip number are investigated. The experimental results indicate that the heat transfer coefficient is strongly affected by Reynolds number and fully-developed values of heat transfer coefficient are reached before the first chip. Empirical correlations are developed for relations using Nusselt number, Reynolds number and Grashof number, based on channel hydraulic diameter.

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Nomenclature

A	exposed chip surface area (m^2)
C	coefficient in correlations
c_p	specific heat at constant pressure (kJ/kg K)
D	channel hydraulic diameter = $2WH/(W + H)$ (m)
Gr_D	Grashof number defined in Eq. (6)
Gr_ℓ	Grashof number defined in Eq. (7)
g	gravitational acceleration (m/s^2)
h	heat transfer coefficient ($\text{W/m}^2 \text{K}$)
k	thermal conductivity of fluid (W/m K)
L	length of the channel (m)
l	length of heat source (m)
m	exponent of (Gr_D/Re_D^2)
n	exponent of Re_D
Nu_ℓ	Nusselt number defined in Eq. (3)
Pe_ℓ	Peclet number defined in Eq. (8)
Pr	Prandtl number = ν/α
Q	flow rate (m^3/s)
q	heater power (W)
q''	heat flux (W/cm^2)
Re_D	Reynolds number defined in Eq. (4)
Re_ℓ	Reynolds number defined in Eq. (5)
S	side wall thickness (m)
SD	standard deviation
T_b	bulk fluid temperature (K) defined in Eq. (1)
T_{in}	channel inlet temperature (K)
T_{wall}	average chip wall temperature (K)
ΔT	temperature difference (K) defined in Eq. (2)
U	velocity of flow = QWH (m/s)
W	width of the channel (m)

Greek symbols

α	thermal diffusivity = $k/(\rho c_p)$ (m^2/s)
β	co-efficient of volumetric thermal expansion ($1/\text{K}$)
μ	dynamic viscosity (kg/ms)
ρ	density of fluid (kg/m^3)
ν	kinematic viscosity (m^2/s)

Subscripts

b	bulk fluid
in	inlet
wall	chip wall

1. Introduction

In the thermal management of high-power electronic chips, liquid cooling has been considered to replace air-cooling, as increasing power density makes it difficult to maintain reasonable chip wall temperature to below 85 °C. Although forced convection has proven to be reliable to remove heat from high-power chips, it necessitates a pump for liquid and fan for air-cooling, and there may be accompanied undesirable noise and vibration. Natural convection, where buoyancy force due to a temperature difference acts on the fluid causing it to flow up through the channel, is more reliable and is low cost. On the other hand, mixed convection may offer the potential for significant cooling enhancement with only modest penalty in system pressure losses, and such flow condition may be well suited to electronics cooling. Under low forced convection conditions, with increase in heat fluxes within the device, the buoyancy effect assumes greater importance, leading to a mixed convection situation.

Braaten and Patankar [1] studied numerically for two-dimensional situation, the mixed convection from a shrouded array of 3×1 in-line heat sources in the horizontal orientation using air, and showed the effects of buoyancy and Prandtl number on the Nusselt number. Türkoğlu and Yücel [2] examined numerically for two-dimensional laminar mixed convection for air-cooling in a vertical channel with a single discrete heat source. The experimental results obtained by Wang and Vafai [3] for the 4×1 in-line in air in a horizontal rectangular channel show that the mixed convection play an important role in the total heat transfer process. Meinders et al. [4] investigated mixed convection experiments for air from a 6×1 in-line protruded elements mounted on the wall of a horizontal wind tunnel. Du et al. [5] numerically simulated for two-dimensional idealization, the mixed convection for air in open-ended vertical channels for various arrangements of protruding heaters.

Based on the survey, it is found that no work has been reported for discrete heat sources in vertical channel liquid cooling under free or mixed convection conditions. Furthermore, none of the above worked on a 4×1 vertical rectangular channel. In the authors' laboratory the forced convection through this geometry has been studied [6,7]. Since this basic arrangement has not been studied for other convection modes, therefore the main contribution in the present work is to cover the three convection modes from a linear array of flush-mounted heat sources in a vertical up-flow channel, using water as the working fluid. The results are compared with the other results from literature. Finally developed an empirical correlation equation to suit for the three convection modes.

2. Model configuration, parameters and operation

The physical model is shown in Fig. 1. The rectangular channel is 840 mm $\times$ 20 mm $\times$ 5 mm and is mounted vertically upwards with coolant being pumped in from the bottom inlet. An array of four in-line square chips (10 mm $\times$ 10 mm) is flush-mounted on one of the channel walls to simulate the discrete heat sources. They are embedded along the channel centre at a spacing of 10 mm and 5 mm between the edges of the chip and the channel side walls, respectively. The chip configuration is numbered from 1 to 4 as the chip elevates vertically upwards.

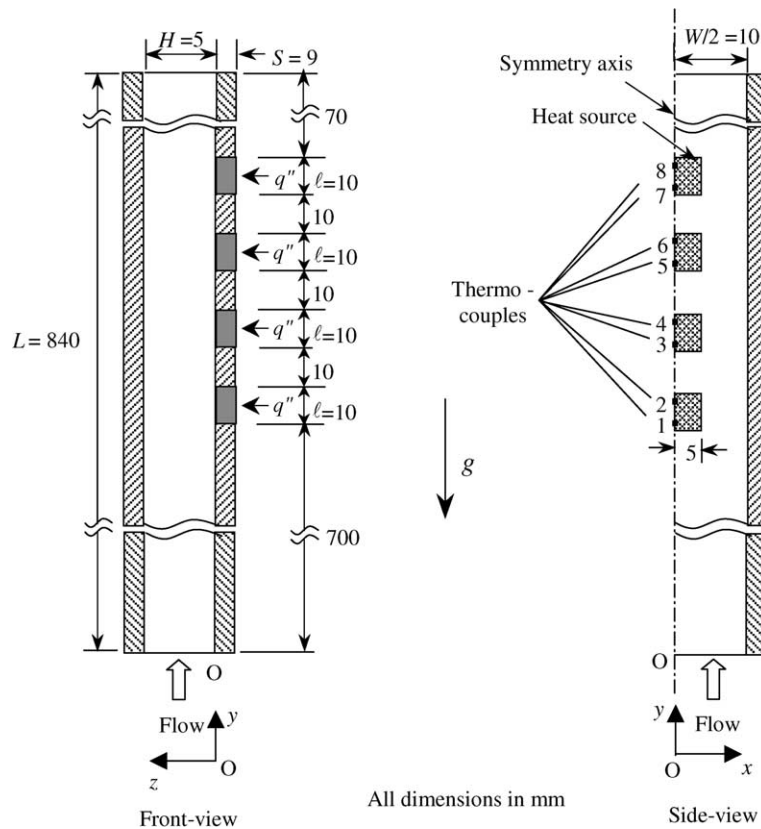


Fig. 1. Schematic diagram of physical model and flow orientation.

Prior to performing the experiments, the chip surfaces are polished with a waterproof abrasive silicon carbide paper that has particles of average size 10 μm and should create a uniform surface texture on each chip. Two chromel-alumel thermocouples of the *K*-type are embedded along the chip centreline in the flow direction at a depth of 0.5 mm under the chip surface and are located 2 mm apart from upstream and downstream edges of the chip, respectively. All experiments with water are performed in a vertical upflow channel and Reynolds number based on channel hydraulic diameter ranging from 40 to 2220 with an inlet temperature of about $T_{\text{in}} \approx 24^\circ\text{C}$. The range of heat flux ranges from 0.1 W/cm^2 to 0.6 W/cm^2 . The configuration of simulated chips tested in this study is always flush-mounted and all the four chips are heated simultaneously at constant heat flux. The chip temperatures always remain less than 70°C .

The heat losses are determined by measuring the surface temperature of Teflon exposed to the surroundings and the temperature of ambient air, and are based on the assumption that the heat losses by conduction in the multichip module are equal to the heat dissipation by natural convection from the surface of the multichip module to the surroundings, the contact resistance between the copper block and the Teflon insulation being negligible. The percentage of heat loss is estimated to be less than 5% of the total heat energy dissipated by the heater. Therefore, in light of the small heat losses, no correction is made for the power dissipated by the thick-film resistor

in determining the chip heat flux. In general, heat losses for liquid cooling of simulated chips have been found to be negligible, such as in the studies of Incropera et al. [8], Willingham and Mudawar [9], and Heidel et al. [10]. An estimation of the overall uncertainty in the experimental data is made using standard techniques for single-sample measurements [11], and the propagation of the uncertainties into dimensionless parameters is then determined. The study reveals uncertainties in ΔT , q'' , Re_ℓ , Re_D , Nu_ℓ , Gr_ℓ , Gr_D , to be less than 1.4%, 5.0%, 3.8%, 3.5%, 6.8%, 3.5% and 3.8%, respectively. These values are based on the assumption of negligible uncertainty in the relevant fluid properties.

3. Analyses of experimental data

All the thermophysical properties [12,13] are evaluated at the bulk fluid inlet temperature. The bulk temperature is defined as

$$T_b = \frac{T_{\text{wall}} + T_{\text{in}}}{2}. \quad (1)$$

The temperature difference between chip and liquid inlet is defined as

$$\Delta T = T_{\text{wall}} - T_{\text{in}}. \quad (2)$$

The data were reduced in terms of Nusselt number based on heat source length as

$$Nu_\ell = \frac{h\ell}{k} = \frac{q\ell}{kA(T_{\text{wall}} - T_{\text{in}})}, \quad (3)$$

where T_{wall} is the average of a chip wall temperature measured by two thermocouples attached on each chip. The chip length is $\ell = 10$ mm, providing exposed chip surface area, $A = 100 \text{ mm}^2$. Reynolds number based on hydraulic diameter, and based on heat source length are defined as

$$Re_D = \frac{\rho U D}{\mu}, \quad (4)$$

$$Re_\ell = \frac{\rho U \ell}{\mu}. \quad (5)$$

The Grashof number based on channel hydraulic diameter, and based on heat source length are defined as

$$Gr_D = \frac{g\beta q'' D^4}{k\nu^2}, \quad (6)$$

$$Gr_\ell = \frac{g\beta q'' \ell^4}{k\nu^2}. \quad (7)$$

The Peclet number based on heat source length, is defined as

$$Pe_\ell = Re_\ell \times Pr. \quad (8)$$

4. Experimental results and analyses

After the fluid flow and the inlet temperature reached steady-state, the electrical power to the chips was turned on. The wall temperature of the simulated chips was allowed to reach steady-state, typically in about 30 min. It was observed that in the last 3 min, the variation of the chip wall temperature less than 0.3 °C. Thus, it can be assumed that the chip wall temperature has reached steady-state at that time and all the readings have been taken at this point. The results obtained from the steady-state were first analysed and compare with the results for forced convection condition with similar geometry and the results are given in Fig. 2. It is found good agreement with the results of Tso et al. [6] and Shah and London [14] for overall data of the four chips that ensured that the test rig is in good working conditions. The analysis is then followed by mixed convection conditions.

Nusselt number as a function of Reynolds number based on heat source length for four chips are shown in Fig. 3. It is seen that as the Reynolds number is decreased, heat transfer enhancement above the forced convection limit initially occurs for the fourth heater. This behavior results from the onset of thermal instabilities and the probable development of secondary flow due to heating at the three upstream heaters. However, the Nusselt number continues to decrease with decreasing Reynolds number until a minimum is reached, where the reduction in Nusselt number of the fourth heater due to forced convection effects is balanced by an increase due to the buoyancy-driven secondary flow. With a further reduction of Reynolds number, the strength of the buoyancy-induced secondary flow increases and the corresponding enhancement more than offsets the decrease due to reduction in the forced flow. Having achieved conditions dominated by mixed convection, a further decrease in Reynolds number results in decay of the forced convection component, transition to pure natural convection with a gradual decreasing of Nusselt number. Although the data appear to be strongly Reynolds number dependent, the results will be shown to correspond to pure natural convection. The relative importance of forced, mixed and free convection is determined by the parameter (Gr_D/Re_D^2) . However it also depends on the Prandtl

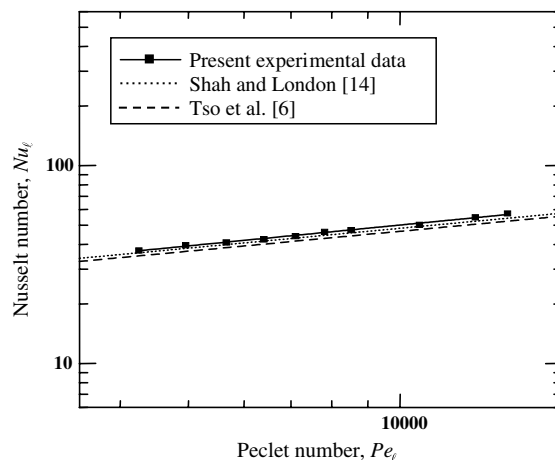


Fig. 2. Nusselt number as a function of Peclet number for overall data.

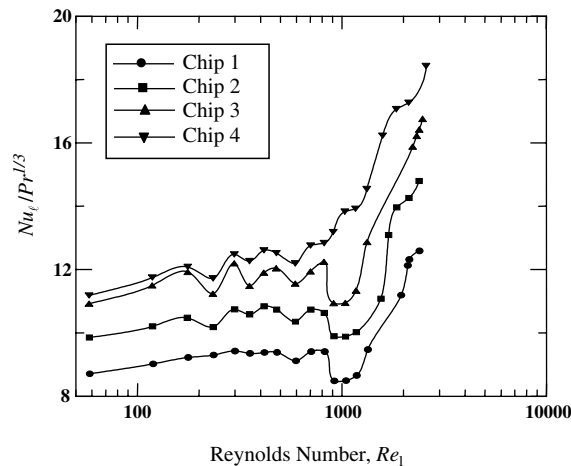


Fig. 3. Variation of $Nu_{\ell}/Pr^{1/3}$ with Reynolds number.

number, which means that for a smaller Prandtl number, even with (Gr_D/Re_D^2) of the order of 1.0, one can observe considerable effect of mixed convection.

It has been proposed that mixed convection occurrence of both natural and forced is to be considered when (Gr_D/Re_D^2) is of the order of unity (Incropera and Dewitt [15]; Lloyd and Sparrow [16]). For $(Gr_D/Re_D^2) \gg 1$, one can assume pure natural convection and for $(Gr_D/Re_D^2) \ll 1$ it shows forced convection behavior clearly. Fig. 4 shows the free and forced convection regions for average of the four chips, the flow rate ranging from $(0.5 \times 10^{-3} \text{ m}^3/\text{h})$ to $20 \times 10^{-3} \text{ m}^3/\text{h}$ and heat flux ranging from $0.1 \text{ W}/\text{cm}^2$ to $0.6 \text{ W}/\text{cm}^2$.

It can be seen that the pure natural convection line merges with the mixed convection solution for $(Gr_D/Re_D^2) \gg 1$; also the pure forced convection merges with the mixed convection solution

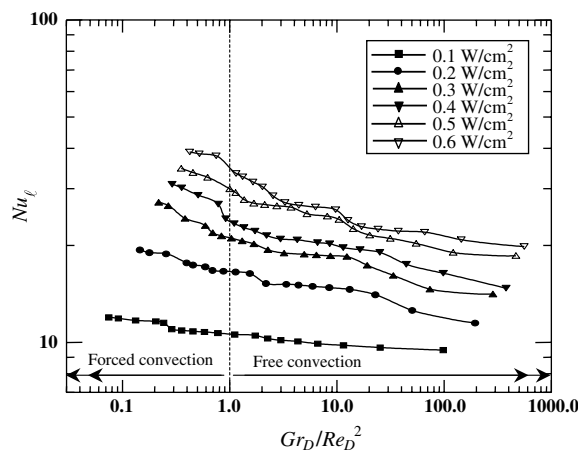


Fig. 4. Variations of Nu_{ℓ} with $[Gr_D/Re_D^2]$ at different heat fluxes.

for $(Gr_D/Re_D^2) \ll 1$. It is seen that from 0.3 W/cm^2 to 0.6 W/cm^2 , there is a clear indication that around unity for (Gr_D/Re_D^2) , there is a change in heat convection mode. The mixed convection region is less clear for 0.1 W/cm^2 and 0.2 W/cm^2 , with 0.1 W/cm^2 having a flat characteristic. The experimental data obtained for all test cases are then used for the development of an empirical correlation for general convection. It is proposed that the empirical correlation for general convection to be the following form:

$$\frac{Nu_\ell}{Re_D^n} = C \left(\frac{Gr_D}{Re_D^2} \right)^m \quad (9)$$

The exponent n of Reynolds number of the left hand side of Eq. (9) has been selected from different test cases with different exponents, starting from 0.5 to 2.5. By choosing an appropriate value of n , it may be possible to obtain a single correlation representing the heat transfer behavior without depending on heat flux. To determine the heat flux independence, the exponent of Re_D of (Nu_ℓ/Re_D) was increased in such way that all the curves are close to each other. The standard deviation (SD) of C and m of Eq. (9) at different values of exponent n of (Nu_ℓ/Re_D^n) are shown in Fig. 5. It is seen that the standard deviation (SD) of C and m are minimum at the unique value, $n = 1.15$. Therefore the exponent for Re_D of (Nu_ℓ/Re_D) is selected as 1.15. The variations of $(Nu_\ell/Re_D^{1.15})$ with (Gr_D/Re_D^2) for all present experimental points for average of the four chips are shown in Fig. 6.

From the survey, it is found that the conditions of Türkoğlu and Yücel [2] are nearer to our experimental conditions and the data collected for mixed convection are plotted in the same scale. For comparison, Grashof number and Reynolds number are re-calculated based on the heat source length as shown in Fig. 7. It is seen that even though the curves are not close to each other, they are of similar slope. In fact we do not expect good agreement with Türkoğlu and Yücel because they are two-dimensional numerical results for single heat source also the different aspect ratio of the channel. However, the heat transfer coefficient is known to be a strong function of aspect ratio and method of heating [17]. It is found that our Nusselt number is about 35% lower than that of the results of Türkoğlu and Yücel.

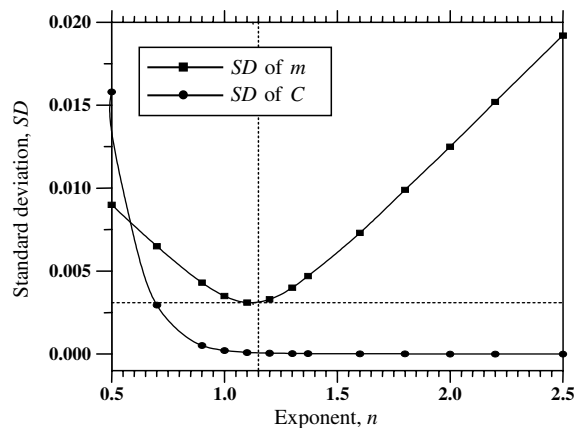


Fig. 5. Variation of exponent n with standard deviation SD.

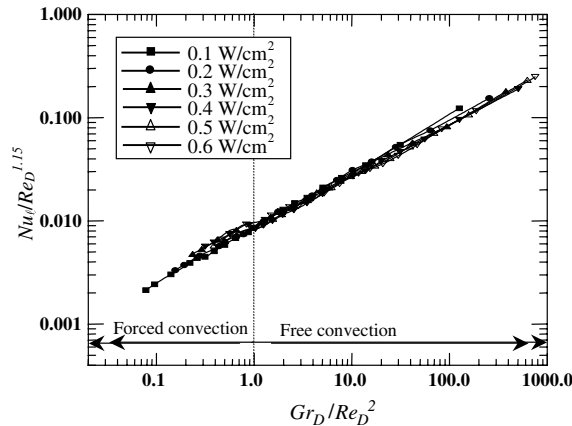


Fig. 6. Variation of $[Nu_\ell/Re_D^{1.15}]$ with $[Gr_D/Re_D^2]$.

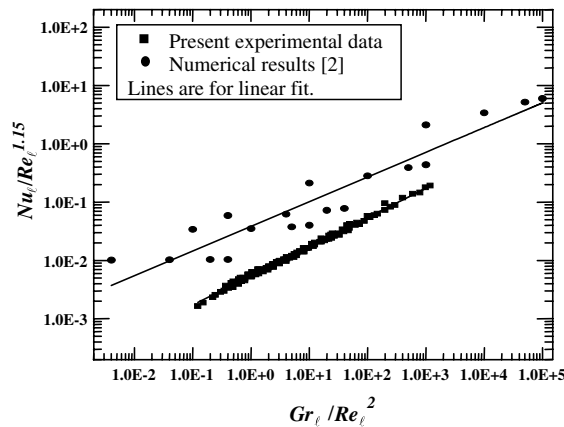


Fig. 7. Variation of $[Nu_\ell/Re_\ell^{1.15}]$ with $[Gr_\ell/Re_\ell^2]$ for average experimental results.

From Fig. 7, it is also found that the results seem to be applicable for all the three convective regimes: free, mixed and forced. The empirical correlation for mixed convection, Eq. (9) can be rewritten in the following form

$$\frac{Nu_\ell}{Re_D^{1.15}} = C \left(\frac{Gr_D}{Re_D^2} \right)^m \quad (10)$$

The C and m in Eq. (10) for the four chips are determined by using the linear fit method to correlate the laminar flow data within the convection range, and results are shown in Fig. 8. The average standard deviation of C and m are less than 21% and 20%, respectively.

From the analysis it is found that there is a higher heat transfer coefficient for the first chip and a drop in heat transfer coefficient as the chip number increases. The Nu_ℓ of the second, third and fourth chips are about 98%, 82% and 78% of the first chip, respectively. Thus, the development of the thermal boundary layer decreases the Nu_ℓ with increasing chip number, until a fully-developed

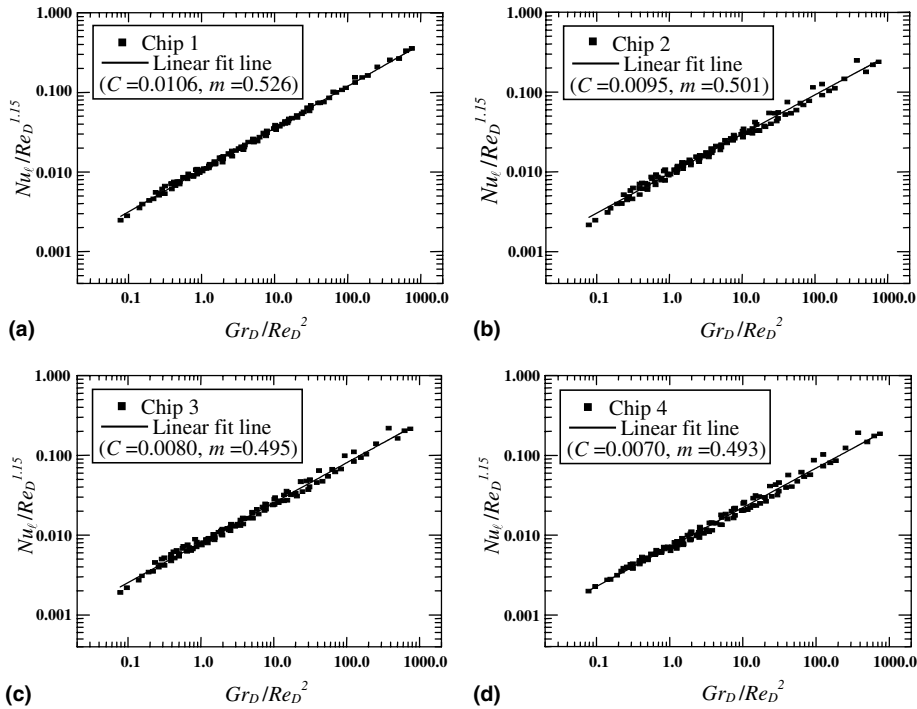


Fig. 8. Linear fit results for each chip. (a) Chip 1, (b) Chip 2, (c) Chip 3, and (d) Chip 4.

condition is reached at approximately the fourth chip. This observation is similar to the forced convection condition [6]. Thus, the overall recommended correlations for the four chips is

$$\frac{Nu_{\ell}}{Re_D^{1.15}} = 0.0088 \left(\frac{Gr_D}{Re_D^2} \right)^{0.5}. \quad (11)$$

5. Conclusions

Experiments have been performed to study the single-phase heat transfer characteristics on an array of four in-line, flush-mounted simulated chips in a vertical up-flow rectangular channel during steady-state operation to determine the overall heat transfer coefficient. The effects of heat fluxes, coolant flow rates and geometrical parameters such as chip configuration numbers are investigated. Finally a single correlation is developed to describe the convection modes.

The experimental results on steady-state forced convection from the chips in the present study agree well with those obtained from the literature from the similar geometry. The experimental results for mixed convection from the present study agree well with the results from the literature for air-cooling of heat sources of similar geometry and the steady-state empirical correlations for general convection have been developed. The overall recommended correlation for the present conditions is given by Eq. (11).

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